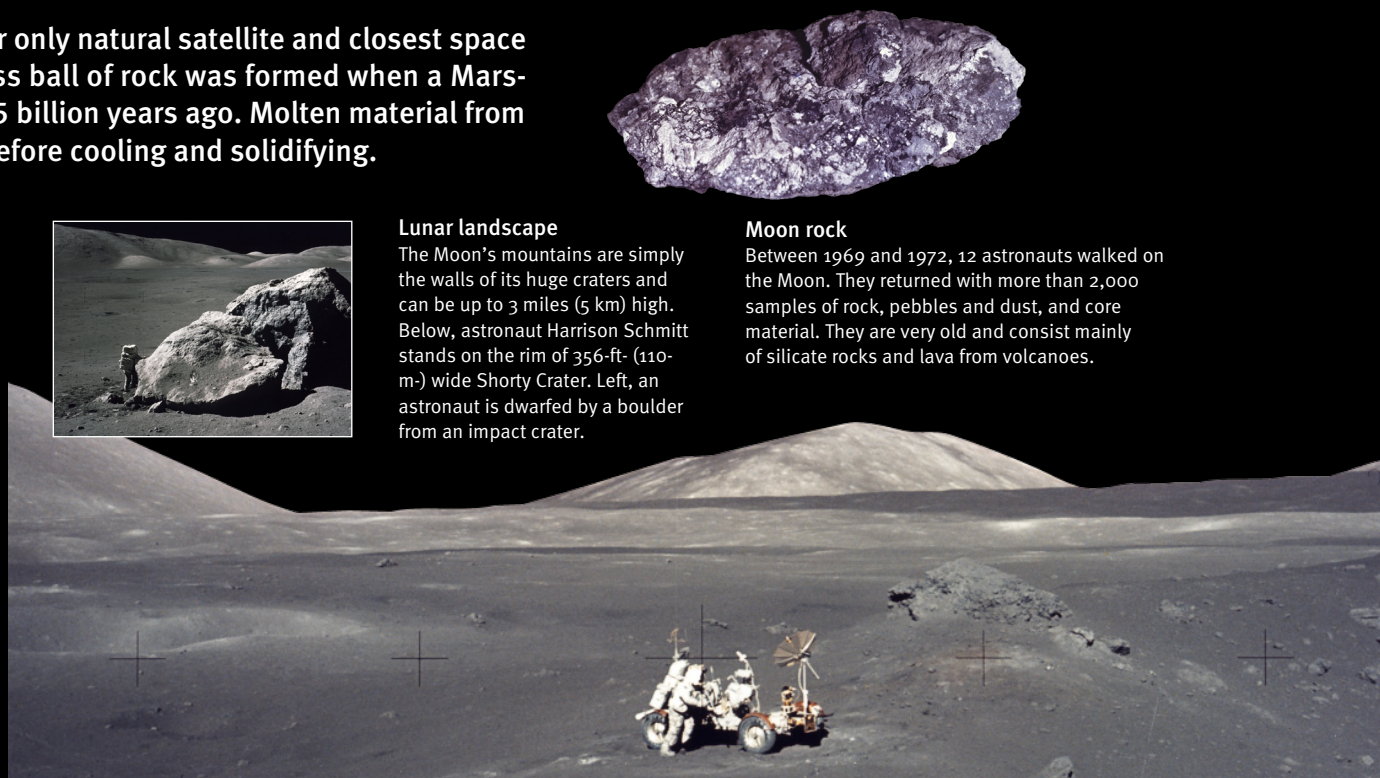


The Moon

The Moon dominates the night sky. It is our only natural satellite and closest space neighbor. Many think this cold, dry, lifeless ball of rock was formed when a Mars-sized asteroid collided with Earth about 4.5 billion years ago. Molten material from Earth and the asteroid formed the Moon, before cooling and solidifying.

FEATURES

The lunar crust is solid and rigid. It is about 30 miles (48 km) thick on the Earth-facing side, and 46 miles (74 km) thick on the far side. Below is a rocky mantle, and because the interior gets hotter with depth, this is partly molten. A small iron core may be at the Moon's center. Bombardments by asteroids and meteorites have pulverized the surface, producing a lunar soil, the regolith, some 16–33 ft (5–10 m) thick. Boulders that have been blasted out of both near and distant craters litter the landscape. The craters, formed by impacting asteroids, range from bowl-shaped ones, less than 6 miles (10 km) across, to features more than 90 miles (150 km) wide. These larger ones have been flooded by lava that seeped from inside the Moon.



Lunar landscape

The Moon's mountains are simply the walls of its huge craters and can be up to 3 miles (5 km) high. Below, astronaut Harrison Schmitt stands on the rim of 356-ft- (110-m-) wide Shorty Crater. Left, an astronaut is dwarfed by a boulder from an impact crater.

Moon rock

Between 1969 and 1972, 12 astronauts walked on the Moon. They returned with more than 2,000 samples of rock, pebbles and dust, and core material. They are very old and consist mainly of silicate rocks and lava from volcanoes.

THE MOON

THE VIEW FROM EARTH

The Moon appears to be the largest object in the night sky. It produces no light of its own but shines by reflecting light from the Sun. From Earth, we only ever see one side of the Moon because the Moon spins around in the same amount of time as it takes to complete one orbit around Earth. The Moon seems to be a different shape from day to day. But these shapes, known as phases, are simply the different amounts of the Moon's sunlit side visible from Earth. A complete cycle of phases lasts 29.5 days.



NORTHERN HEMISPHERE



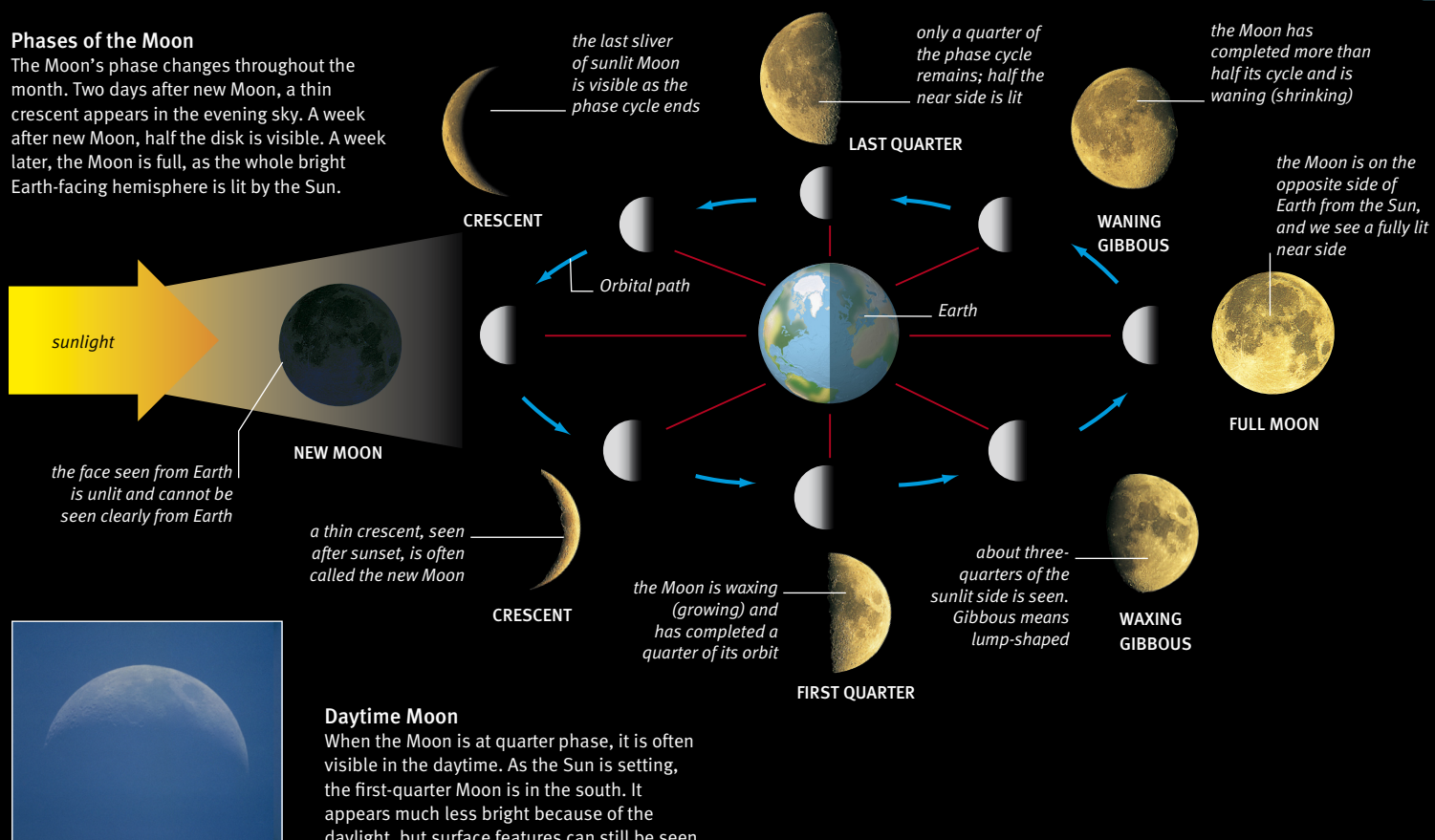
SOUTHERN HEMISPHERE

Changing views

The European astronomers who first mapped the Moon called the top "north" and the bottom "south." But southern-hemisphere observers see the Moon's south pole at the top.

Phases of the Moon

The Moon's phase changes throughout the month. Two days after new Moon, a thin crescent appears in the evening sky. A week after new Moon, half the disk is visible. A week later, the Moon is full, as the whole bright Earth-facing hemisphere is lit by the Sun.



Daytime Moon

When the Moon is at quarter phase, it is often visible in the daytime. As the Sun is setting, the first-quarter Moon is in the south. It appears much less bright because of the daylight, but surface features can still be seen.